

MA1521 L1 Week 05 Mon 20260907

Find the differential (dw or dz). Make the answer look as neat as possible.

a) $w = \ln(xyz)$ b) $w = x^3y^2z$ c) $z = \frac{x-y}{x+y}$

$$\begin{aligned} dw &= \frac{d(xyz)}{xyz} \\ &= \frac{yz \, dx + xz \, dy + xy \, dz}{xyz} \\ &= \frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} \end{aligned}$$

$$\begin{aligned} dw &= d(x^3y^2z) \\ &= 3x^2y^2z \, dx \\ &\quad + 2x^3y \, dy \\ &\quad + x^3y^2 \, dz \end{aligned}$$

$$\begin{aligned} dz &= d\left(\frac{x-y}{x+y}\right) \\ &= \frac{(x+y) \, d(x-y) - (x-y) \, d(x+y)}{(x+y)^2} \\ &= \frac{1}{(x+y)^2} \left[(x+y)(dx - dy) - (x-y)(dx + dy) \right] \\ &= \frac{2y \, dx - 2x \, dy}{(x+y)^2} \end{aligned}$$

quot. rule

The following equations define w implicitly as a function of the other variables. Find dw in terms of all the variables by taking the differential of both sides and solving algebraically for dw .

$$\text{a) } \frac{1}{w} = \frac{1}{t} + \frac{1}{u} + \frac{1}{v} \quad \text{b) } u^2 + 2v^2 + 3w^2 = 10$$

Apply "d" to $\frac{1}{w} = \frac{1}{t} + \frac{1}{u} + \frac{1}{v}$

get $d\left(\frac{1}{w}\right) = d\left(\frac{1}{t} + \frac{1}{u} + \frac{1}{v}\right)$

$$\frac{-1}{w^2} dw = \frac{-1}{t^2} dt + \frac{-1}{u^2} du + \frac{-1}{v^2} dv$$

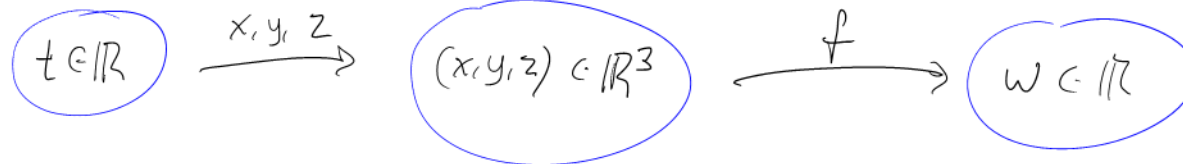
$$dw = w^2 \left(\frac{dt}{t^2} + \frac{du}{u^2} + \frac{dv}{v^2} \right)$$

Apply "d" to $u^2 + 2v^2 + 3w^2 = 10$

get $d(u^2 + 2v^2 + 3w^2) = d(10) = 0$

i.e. $2u du + 4v dv + 6w dw = 0$

$$dw = \frac{-1}{3w} (u du + 2v dv)$$



$$w = f(x, y, z) \\ = \underbrace{f(x(t), y(t), z(t))}_{\text{composite}}$$

In the following, find $\frac{df}{dt}$ for the composite function $f(x(t), y(t), z(t))$ in two distinct ways:

- use the chain rule, then express your answer in terms of t by using $x = x(t)$, etc.;
- express the composite function f in terms of t , and differentiate.

a) $w = xyz$, $x = t$, $y = t^2$, $z = t^3$

b) $w = x^2 - y^2$, $x = \cos t$, $y = \sin t$

c) $w = \ln(u^2 + v^2)$, $u = 2 \cos t$, $v = 2 \sin t$

$$\begin{aligned} \text{a) (i)} \quad \frac{dw}{dt} &= \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{pmatrix} \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \\ \frac{dz}{dt} \end{pmatrix} = \underbrace{\frac{\partial f}{\partial x}}_{yz} \underbrace{\frac{dx}{dt}}_1 + \underbrace{\frac{\partial f}{\partial y}}_{xz} \underbrace{\frac{dy}{dt}}_{2t} + \underbrace{\frac{\partial f}{\partial z}}_{xy} \underbrace{\frac{dz}{dt}}_{3t^2} \\ &= 6t^5 \quad \text{using } \underline{\text{Chain rule}} \end{aligned}$$

$$f(x, y, z) = xyz$$

$$\begin{aligned} x(t) &= t \\ y(t) &= t^2 \\ z(t) &= t^3 \end{aligned}$$

(ii) directly, get $w(t) = \underbrace{x(t)}_t \underbrace{y(t)}_{t^2} \underbrace{z(t)}_{t^3} = t^6$

so $\frac{dw}{dt} = 6t^5$

b) (i) $w = x^2 - y^2, \quad x = \cos t, \quad y = \sin t$

chain rule

$$\frac{dw}{dt} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \underbrace{\frac{\partial f}{\partial x}}_{2x} \underbrace{\frac{dx}{dt}}_{-\sin t} + \underbrace{\frac{\partial f}{\partial y}}_{-2y} \underbrace{\frac{dy}{dt}}_{\cos t}$$

$$= -4 \sin t \cos t$$

$$(= -2 \sin 2t)$$

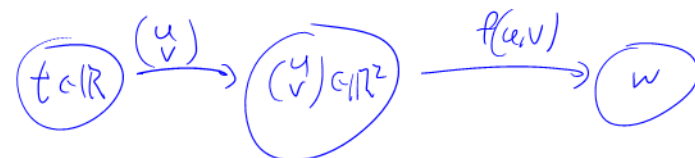
$$f(x, y) = x^2 - y^2$$

$$x^{(t)} = \cos t$$

$$y^{(t)} = \sin t$$

ii) directly: $w(t) = x(t)^2 - y(t)^2 = \cos^2 t - \sin^2 t = \cos 2t$

so $\frac{dw}{dt} = -2 \sin 2t$



$$u(t) = 2 \cos t$$

$$v(t) = 2 \sin t$$

$$f(u, v) = \ln(u^2 + v^2)$$

c) (i) $w = \ln(u^2 + v^2), \quad u = 2 \cos t, \quad v = 2 \sin t$

Chain rule.

$$\frac{dw}{dt} = \begin{pmatrix} \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} \end{pmatrix} \begin{pmatrix} \frac{du}{dt} \\ \frac{dv}{dt} \end{pmatrix} = \underbrace{\frac{\partial f}{\partial u}}_{\frac{2u}{u^2+v^2}} \underbrace{\frac{du}{dt}}_{-2 \sin t} + \underbrace{\frac{\partial f}{\partial v}}_{\frac{2v}{u^2+v^2}} \underbrace{\frac{dv}{dt}}_{2 \cos t}$$

$$= 0$$

directly: $w(t) = \ln(u(t)^2 + v(t)^2) = \ln(4 \cos^2 t + 4 \sin^2 t) = \ln 4 \text{ const!}$

$$\frac{dw}{dt} = 0$$

a) Use the chain rule for $f(u, v)$, where $u = u(t)$, $v = v(t)$, to prove the product rule

$$D(uv) = v Du + u Dv, \quad \text{where } D = \frac{d}{dt}.$$

b) Using the chain rule for $f(u, v, w)$, derive a similar product rule for $\frac{d}{dt}(uvw)$, and use it to differentiate $te^{2t} \sin t$.

$$\begin{aligned} \text{a) } \frac{d}{dt} \underbrace{(u(t) v(t))}_{f(u, v)} &= \begin{pmatrix} f_u & f_v \end{pmatrix} \begin{pmatrix} u'(t) \\ v'(t) \end{pmatrix} \\ &= \underbrace{\frac{\partial f}{\partial u}}_v \cdot \frac{du}{dt} + \underbrace{\frac{\partial f}{\partial v}}_u \cdot \frac{dv}{dt} = v \cdot \frac{du}{dt} + u \cdot \frac{dv}{dt} \end{aligned}$$

$$\begin{array}{ccc} t \in \mathbb{R} & \longrightarrow & (u, v) \in \mathbb{R}^2 \xrightarrow{f} w \in \mathbb{R} \\ & & (u(t), v(t)) \quad \underline{f(u, v) = uv} \end{array}$$

$$\text{b) } \frac{d}{dt} \underbrace{(u(t) v(t) w(t))}_{f(u, v, w)}$$

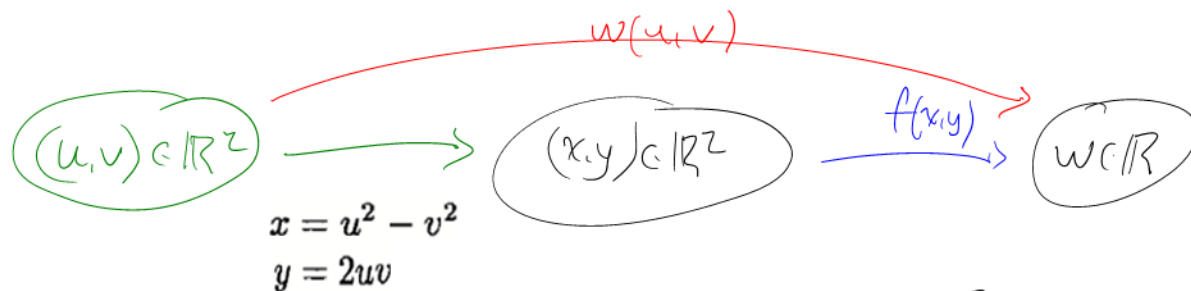
$$\begin{array}{ccc} t \in \mathbb{R} & \longrightarrow & (u, v, w) \in \mathbb{R}^3 \xrightarrow{f} w \in \mathbb{R} \\ & & (u(t), v(t), w(t)) \quad \underline{f(u, v) = uvw} \end{array}$$

$$\begin{aligned} &= \begin{pmatrix} f_u & f_v & f_w \end{pmatrix} \begin{pmatrix} u'(t) \\ v'(t) \\ w'(t) \end{pmatrix} = \underbrace{\frac{\partial f}{\partial u}}_{vw} \cdot \frac{du}{dt} + \underbrace{\frac{\partial f}{\partial v}}_{uw} \cdot \frac{dv}{dt} + \underbrace{\frac{\partial f}{\partial w}}_{uv} \cdot \frac{dw}{dt} = vw \frac{du}{dt} + uw \frac{dv}{dt} + uv \frac{dw}{dt} \end{aligned}$$

$$\begin{aligned} \frac{d}{dt} \underbrace{(te^{2t} \sin t)}_{\substack{\uparrow u \quad \uparrow v \quad \uparrow w}} &= e^{2t} \sin t \cdot \frac{dt}{dt} + t \sin t \cdot \frac{d(e^{2t})}{dt} + t e^{2t} \cdot \frac{d(\sin t)}{dt} \\ &= e^{2t} \sin t + t \sin t \cdot 2e^{2t} + t e^{2t} \cdot \cos t = e^{2t} (\sin t + 2t \sin t + t \cos t) \end{aligned}$$

Let $w = f(x, y)$, and make the change of variables $x = u^2 - v^2$, $y = 2uv$. Show

$$(w_x)^2 + (w_y)^2 = \frac{(w_u)^2 + (w_v)^2}{4(u^2 + v^2)}$$



By chain rule, $\begin{pmatrix} \frac{\partial w}{\partial u} & \frac{\partial w}{\partial v} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix} \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$

i.e. $w_u = f_x \cdot \frac{x_u}{2u} + f_y \cdot \frac{y_u}{2v} = 2uf_x + 2vf_y$

$w_v = f_x \cdot \frac{x_v}{-2v} + f_y \cdot \frac{y_v}{2u} = -2vf_x + 2uf_y$ using $\begin{cases} x = u^2 - v^2 \\ y = 2uv \end{cases}$

hence $\frac{(w_u)^2 + (w_v)^2}{4(u^2 + v^2)}$

$\nearrow = w_x^2 + w_y^2$

so $(w_u)^2 + (w_v)^2 = (2uf_x + 2vf_y)^2 + (-2vf_x + 2uf_y)^2$
 $= 4u^2 f_x^2 + 4v^2 f_y^2 + 8uv f_x f_y = 4(u^2 + v^2)(\underbrace{f_x^2}_{w_x^2} + \underbrace{f_y^2}_{w_y^2})$
 $+ 4v^2 f_x^2 + 4u^2 f_y^2 - 8uv f_x f_y$

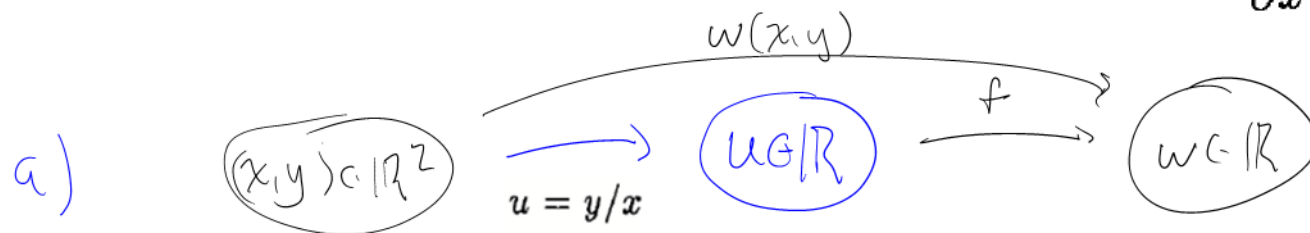
a) Let $w = f(y/x)$; i.e., w is the composite of the functions $w = f(u)$, $u = y/x$.

Show that w satisfies the PDE (partial differential equation)

$$x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} = 0$$

b) Let $w = f(x^2 - y^2)$; show that w satisfies the PDE $y \frac{\partial w}{\partial x} + x \frac{\partial w}{\partial y} = 0$.

c) Let $w = f(ax + by)$; show that w satisfies the PDE $b \frac{\partial w}{\partial x} - a \frac{\partial w}{\partial y} = 0$.



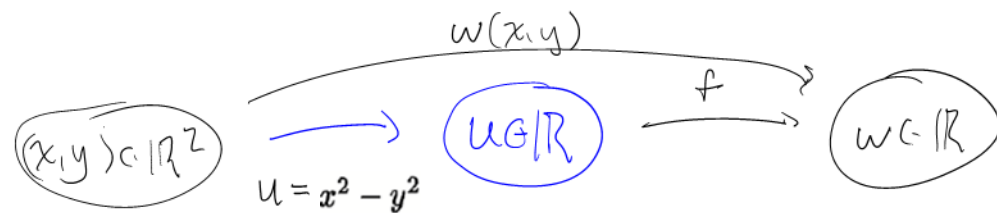
Chain rule: $\begin{pmatrix} \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{df}{du} \end{pmatrix} \cdot \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \end{pmatrix}$

i.e., $\frac{\partial w}{\partial x} = \frac{df}{du} \left(\frac{\partial u}{\partial x} \right) = \frac{df}{du} \cdot \left(-\frac{y}{x^2} \right)$

$\frac{\partial w}{\partial y} = \frac{df}{du} \left(\frac{\partial u}{\partial y} \right) = \frac{df}{du} \cdot \left(\frac{1}{x} \right)$ using $u = y/x$

Hence $x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} = x \cdot \frac{df}{du} \cdot \left(-\frac{y}{x^2} \right) + y \cdot \frac{df}{du} \cdot \left(\frac{1}{x} \right) = 0$

b)



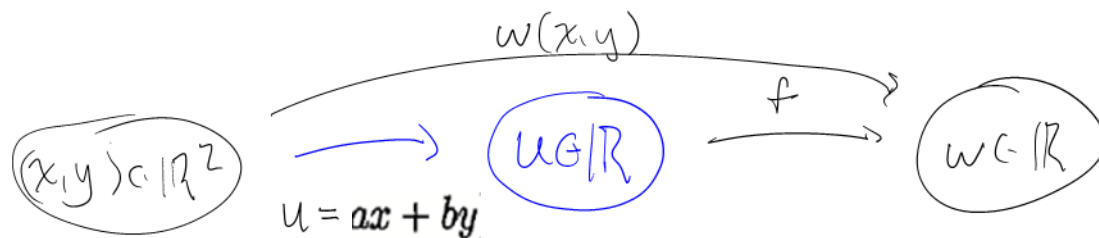
Chain rule: $\begin{pmatrix} \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{pmatrix} = \left(\frac{df}{du} \right) \cdot \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \end{pmatrix}$

i.e., $\frac{\partial w}{\partial x} = \frac{df}{du} \left(\frac{\partial u}{\partial x} \right) = \frac{df}{du} \cdot (2x)$
using $u = x^2 - y^2$

$\frac{\partial w}{\partial y} = \frac{df}{du} \left(\frac{\partial u}{\partial y} \right) = \frac{df}{du} \cdot (-2y)$

Hence $y \frac{\partial w}{\partial x} + x \frac{\partial w}{\partial y}$
 $= y \cdot \frac{df}{du} \cdot (2x) + x \cdot \frac{df}{du} \cdot (-2y)$
 $= 0$

c)



Chain rule: $\begin{pmatrix} \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{pmatrix} = \left(\frac{df}{du} \right) \cdot \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \end{pmatrix}$

i.e., $\frac{\partial w}{\partial x} = \frac{df}{du} \left(\frac{\partial u}{\partial x} \right) = \frac{df}{du} \cdot a$
using $u = ax + by$

$\frac{\partial w}{\partial y} = \frac{df}{du} \left(\frac{\partial u}{\partial y} \right) = \frac{df}{du} \cdot b$

Hence $b \frac{\partial w}{\partial x} - a \frac{\partial w}{\partial y} = b \cdot \frac{df}{du} \cdot a - a \cdot \frac{df}{du} \cdot b = 0$